The location of the exceptional exit in a later shared crossing

Human Compression project: L044 continuation

Retain the complete standing assumptions and smaller-degree alternating induction of *Completing the transitive-tail alternative* and *Routing a maximal shared crossing*. In particular $G \le A_n$ is transitive with a connected string C-marking, every maximal omission is intransitive, and eligibility is always ambient. All full predecessor proofs and their required finite inputs remain acquired dependencies.

Lemma 1 (The two exits have different index locations)**.** *In the exact transitive-tail interface, let i be the unique crossing, $h > i$ its last shared label and $G_{>h}$ transitive on its support. Under rank failure, either the actual last three generators form a six-point Table 2 head, with $h = r-4$, or $h = 2$ and the actual first three generators form such a head. In the latter case $i = 0$ or 1.*

In the fully reversed interface, let j be the crossing and $g < j$ its first shared label, with the prefix $G_{<g}$ transitive on its support. The alternatives are an actual first head with $g = 3$, or an actual last head with $g = r-3$.

Proof. We inspect the full proof, not just its weaker either-end statement. Its small transitive tail is either the degree-five dihedral group, which the exact critical-tail lemma excludes, or an actual six-point A_5 marking. The latter has rank three, so $r-1-h = 3$. The strict proper primitive and anchored-imprimitive bounds exclude all other proper tails.

The remaining natural alternating tail has the precise equality parameters in equation (2) of the acquired proof. In its endpoint case the exact fixed-path lemma forces $i = 0, h = 2$ and the first head. Its transitive-left interior case is impossible. Its intransitive-left interior case forces $h = i+1 = 2$; the complete six-choice dihedral attachment then gives the first head. There are no other exits from that proof. Replacing every index k by $r-1-k$ gives the asserted prefix alternatives: the reversed last shared index is $r-1-g$, so its values $r-4$ and 2 become $g = 3$ and $g = r-3$, respectively. This is full string reversal, not a local reorder of the marked tuple. □

Theorem 2 (A shared crossing beyond the first exceptional head)**.** *Assume an actual extendable six-point Table 2 head at the first three labels and rank failure, so $n = 2r \ge 18$ under the complete critical-head reduction. Among all unique crossings with a shared label to their right choose an index i maximal, and let h be its last shared label. If $h \ge 3$, then either the actual last three generators form a six-point Table 2 head or G_{r-1} has a singleton orbit.*

Proof. Suppose neither conclusion holds. We follow the entire acquired maximal-crossing construction, recording all places where its former exclusion of *both* exceptional ends must be replaced. Its interval-choice argument and the transitive/intransitive support bounds give the actual orientation and

$$h < r-1, \qquad X = \operatorname{supp}(G_{>h}), \qquad n - |X| \ge 2h+1. \tag{1}$$

If $G_{>h}$ were transitive on X, Lemma 1 would force a last head: its first-head exit has $h = 2$, contrary to the hypothesis. Hence this tail is intransitive on its support. The full ambient terminal inequality and (1) give a smallest unique crossing $j > h$, just as in the acquired proof; otherwise $2(r-1-h) \le |X| - 2$ would contradict rank failure.

The two actual bridges yield the same regions A, M, D, with the j-crossing separating $C = A \cup M$ from D. If D is a singleton, the complete interior-singleton bound makes $j = r - 1$, contrary to our exclusion. Assume D nontrivial.

If this later cut is unshared, its actual sides are the supported transitive prefix $G_{<j}$ and suffix $G_{>j}$. Here $j \geq 4$, and $j = 4$ is excluded by the complete early exceptional-cut lemma: in the critical head G_4 has at least three orbits. Thus $j \geq 5$. The complete later-unshared-cut theorem in *Opposite exceptional ends after a later unshared cut* applies. Its only residual alternatives are the terminal singleton or a degree-five dihedral suffix whose full attachment creates an actual last head. Both contradict our exclusions. This step uses its complete small proper-prefix exclusions and the actual natural prefix conclusion, not raw Table 2 or a quotient IP.

There is therefore a first shared label g at j. Maximality of i places it to the left. The entire orientation argument of the acquired maximal-crossing proof gives

$$Y = \operatorname{supp}(G_{<g}), \qquad n - |Y| \geq 2(r - 1 - g) + 1. \tag{2}$$

Its nested-bridge count first proves $g \geq h$ and the requisite purity of the two bridge involutions; the complete touching exclusions then give $g > h$. None of these steps uses absence of the first head.

If $G_{<g}$ were transitive on Y, apply Lemma 1 in this proved reversed orientation. Its first-head exit requires $g = 3$, impossible because $g > h \geq 3$. Its other exit is the excluded last head. Thus both $G_{<g}$ and $G_{>h}$ are intransitive on their actual supports. Every colour strictly between h and g has two ambient-eligible edges, by minimality of $j > h$ and $g < j$. The full orbit-incidence inequality with (1)–(2) now gives $2r \leq n - 1$, the same final contradiction as the acquired proof. These are exactly its three exceptional exits, each replaced without assuming the already present first head absent. □

Exact residual and source scope. This sharpens the either-end alternative supporting source Proposition 5.28, physical pages 36–37, only in the displayed critical first-head situation with $h \geq 3$. The complete early omission calculation shows that $h = 2$ can occur here only for $i = 0$ in the extendable Table 2(2) orientation: colour 1 has two eligible edges, while the Table 2(3) initial unique crossing has no shared labels. If any unique crossing at index at least 5 has a right-shared label, the maximal choice in Theorem 2 automatically has $h \geq 6$. The first-head case with no such later right-shared crossing remains unresolved by this note. So do elimination of the two possible opposite ends, the stronger source rigidity/table contracts and complete L044 recovery. No new computation, formal theorem, original credit or question claim is made.